INCH-POUND

MIL-L-81293A(OS)
22 July 1992
SUPERSEDING
MIL-L-81293(WP)
16 June 1965

MILITARY SPECIFICATION

LIGHTWEIGHT SANDWICH-CONSTRUCTED PANELS (for ASROC Launcher MK 112)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight sandwich-constructed panels for use in the ASROC Launcher MK 112.

1.2 Types. Panels shall be of the following types as specified (see 6.2):

Type I - Side Panel Assembly

Type II - Lower Panel Assembly

Type III - Intermediate Panel Assembly

Type IV - Top Panel Assembly

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 0343), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5640

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

MMM-A-132	Adhesives, Heat Resistant, Airframe, Structural, Metal-to-Metal
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-A-9067	Adhesive Bonding, Process and Inspection Requirements for
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible
MIL-A-25463	Adhesive, Film Form, Metallic Structural Sandwich Construction

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-401	Sandwich Constructions and Core Materials; General Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

5979194 Panel Assembly, Side

5979195 Panel Assembly, Lower

5979196 Panel Assembly, Intermediate

5979197 Panel Assembly, Top

(Application for copies of NAVSEA drawings should be addressed to the Commanding Officer, Naval Ordnance Station, Crane Division, Naval Surface Warfare Center, Attn: Code 802, Louisville, KY 40214-5001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI 1015-1025	Steel

(Application for copies may be obtained from the American Iron and Steel Institute, 1000 16th Street N. W., Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A6	Rolled Steel Plates, Sheet Piling, and Bars for Structural Use (DOD adopted)
ASTM-A36/A36M	Structural Steel (DOD adopted)
ASTM-A108	Steel Bars, Carbon, Cold-Finished, Standard Quality (DOD adopted)

(Application for copies may be otbained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample, consisting of one Intermediate Panel Assembly in accordance with NAVSEA Drawing 5979196, shall be subjected to first article inspection (see 6.3) in accordance with 4.4. The panel shall be laid up and bonded utilizing tooling, materials, and manufacturing methods for regular production runs.

3.2 Design. The panels covered by this specification shall be formed in a three piece laminated construction. The panels shall be formed by bonding ribbon foil between two pieces of sheet aluminum, using a resin as a bonding agent, and applying pressure to position the laminations until the resin is set.

3.2.1 Dimensions. The completed panels shall be types I, II, III or IV conforming to NAVSEA Drawings 5979194, 5779195, 5979196, or 5979197 respectively (see 1.2) as specified in the contract or purchase order (see 6.2). The panels shall meet the dimensions and tolerances specified on the applicable product drawing.

3.3 Materials. Unless otherwise specified in the contract or purchase order (see 6.2), all materials entering into the final assembly of the product shall conform to the requirements indicated on the applicable drawings listed in 3.2.1.

3.4 Finishes and coatings. Finishes and coatings shall be in accordance with the applicable drawings and specifications. It is to be noted that any coating applicable to the finished panel assemblies shall be applied only by brushing, spraying, roller-coating or swabbing. Immersion of bonded panel assemblies in any type solution or fluid connected with the application of a coating is prohibited.

3.5 Marking of assemblies. All completed panels shall be marked in accordance with applicable drawings.

3.6 Product characteristics. The panels shall meet the product characteristics of 3.6.1 through 3.6.8.

3.6.1 Heat resistance. The panels shall withstand, without evidence of cracking, crazing, delamination, or permanent deformation (see 6.4.1), a temperature of 168 +0, -10 degrees Fahrenheit (°F) when undergoing the shear strength and flexural strength tests in 4.6.3 and 4.6.4.

3.6.2 Shear strength. The panels shall withstand, without evidence of failure or permanent deformation, a force of 1680 +40, -0 pounds when tested as specified in 4.6.3.

3.6.3 Flexural strength. The panels shall withstand, without evidence of failure or permanent deformation, the simultaneous application of a 1320 +30, -0 pound force (660 +15, -0 pounds each support) when tested as specified in 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

3.6.4 Density. The panels shall have a weight of not less than 4.10 and not more than 4.36 pounds per square foot.

3.6.5 Lap shear strength. The panels shall have an average lap shear strength of 2500 pounds per square inch (psi) when tested as specified in 4.6.6.

3.6.6 Short column compression strength. The panels shall sustain a load average of 80,000 psi (30,500 pounds actual load) with no two specimens less than 70,000 psi (28,500 pounds actual load) when tested as specified in 4.6.7.

3.6.7 Side panel assembly strength requirement. The completed side panel assembly, NAVSEA Drawing 5979194, shall withstand, without evidence of failure or permanent deformation, a force (p) of 25,800 ± 500 pounds, applied in the manner shown on figure 4. The panel shall show no evidence of failure or permanent deformation during and after testing in accordance with 4.5.4.2.

3.6.8 Top, intermediate, and lower panel assemblies strength requirements. The completed top, intermediate, and lower panel assemblies, NAVSEA Drawings 5979195, 5979196 and 5979197, shall withstand, without evidence of failure or permanent deformation, a force of 12,000 ± 500 pounds, applied in the manner shown on figure 4 when tested in accordance with 4.5.4.3.

3.7 Workmanship. The panels and component parts shall be constructed and finished in such a manner as to assure compliance with all requirements of this specification. Particular attention shall be paid to removal of any adhesive from the outer surface and chemical film coating of the completed panel, neatness and thoroughness of identification marking, and freedom from burrs or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 **Test equipment.** The test equipment required for performing first article and quality conformance inspection shall consist of the following:

a. Two supporting pieces, 1.0 inch by 2.0 inches by 3.0 inches long, made of mild steel conforming to AISI 1015-1025 (Alternate: ASTM-A108, FS 1015-1025). For details of contoured edges, refer to figure 1; pieces are identified by letter A.

b. One pressure piece, 1.5 inches by 2.0 inches by 3.0 inches long, made of mild steel conforming to AISI 1015-1025 (Alternate: ASTM-A108, FS 1015-1025). For details of contoured edges, refer to figure 1; piece is identified by letter B.

c. Four pieces, 0.5 inch by 3.0 inch by 3.0 inch angle, approximately 3.0 inches long, of mild steel conforming to ASTM-A6 (Alternate: ASTM-A36/A36M). These pieces are designated by letter C as shown on figure 2.

d. Four pieces, 1.0 inch by 2.0 inches wide by approximately 3.0 inches long, of mild steel conforming to AISI 1015-1025 (Alternate: ASTM-A108, FS 1015-1025). Each of these pieces, designated by letter D on figure 2, shall have a groove milled on 0.28 inch radius, centered and running longitudinally on the wide face.

e. One piece, 1.0 inch thick by 3.0 inches wide by approximately 22 inches long, of mild steel conforming to AISI 1015-1025 (Alternate: ASTM-A108, FS 1015-1025). Piece is identified by letter F on figure 2. Two of the angles, designated by C, may be welded to piece marked F; the center-to-center distance of the extending legs of the angles shall be 15.0 ± 0.12 inches.

f. Three pieces, 4.0 inches thick by 6.0 inches wide by 64.0 inches long, of mild steel conforming to AISI 1015-1025 (Alternate: ASTM-A108, FS 1015-1025). These pieces are designated as G on figure 4. The edges of these pieces which come in contact with the panel during testing shall have a radius of 0.06 inch.

4.4 **First article inspection.** The panel assembly specified in 3.1 shall be subjected to the first article inspection. The panel shall first be subjected to the applicable strength testing as specified under 4.5.4.3. Upon completion of the strength test, the panel shall be cut to provide test specimens and tested as specified under 4.6.1 through 4.6.7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

Failure of the panel assembly or any of the test specimens cut from the panel assembly to pass the inspection and tests described herein shall be cause for rejection of the first article.

4.5 **Quality conformance inspection.** For the purpose of this specification, two methods of quality conformance inspection are applicable for final tests and examinations on panels fabricated in accordance with 3.2. The methods are:

a. Method A, Nondestructive Testing (see 4.5.4).

b. Method B, Destructive Testing (see 4.5.5).

Method A shall be performed unless there is a change in manufacturing processes during production of the lot.

4.5.1 **Inspection lot.** The inspection lot shall consist of all panels fabricated in a continuous daily production run, not to exceed 40 panels, without a change in process or tooling. Changes in process, tooling, or shutdowns in which production is interrupted for two hours or more, shall constitute the end of one production run and the beginning of a new production run. If required (see 6.2), identification of inspection lots and panels within inspection lots shall be as specified in the contract or purchase order.

4.5.2 **Visual examination.** All panels offered for acceptance will be visually examined for defects in workmanship, and measurements shall be taken to assure that the panels conform with the dimensional requirements of the applicable drawings. Any panels having defects shall be rejected or returned to the contractor for rework. Any panel offered for quality conformance inspection after having been reworked shall be subjected to reexamination and shall undergo Method A, Nondestructive Method test described in 4.5.4. A reworked panel shall not be counted as a portion of any inspection lot as defined in 4.5.1.

4.5.3 **Disposition of rejects.** When the inspection lot is rejected under the provisions of either Method A or B, each panel of the inspection lot shall be subjected to the applicable test of Method A. For acceptance, each panel shall meet the requirements specified. Panels failing to meet the requirements shall be rejected. Rework shall not be permitted on any panel failing to meet the strength requirements when tested in accordance with Method A.

4.5.4 **Method A, nondestructive method.** The lot shall be sampled, tested and accepted in accordance with 4.5.4.1 through 4.5.4.4.

4.5.4.1 **Selection of sample.** One panel shall be selected at random from the inspection lot. The panel may be any of the four panel types covered by the applicable drawings (listed in 3.2.1), when the inspection lot is a mixed production run. The type I panel shall be subjected to the test of 4.5.4.2, and the types II, III, and IV panels, subjected to the test of 4.5.4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

4.5.4.2 Test procedure, side panel. When a side panel, Drawing 5979194, is selected for testing, it shall be laid up and supported as shown on figure 4. The supports shall be 4.0 inches (nominal) wide, and long enough to span the panel. The edges which come in contact with the panel during testing shall have a radius of 0.06 inch. The supports shall be set at 2.0 feet, plus or minus 2.0 inches, center-to-center distance. The pressure head shall be 4.0 inches (nominal) wide, and long enough to span the panel. The edges which come in contact with the panel during testing shall have a radius of 0.06 inch. The load of 25,800 ± 500 pounds shall be applied in eight, approximately equal, successive increments for the entire length of the panel as indicated on figure 4. The rate of load application shall be such that maximum load will occur in not less than three minutes nor more than five minutes. The maximum load shall be maintained for a period of not less than five minutes. The panel shall meet the requirements of 3.6.7 and shall be thoroughly suitable for the intended use.

4.5.4.3 Test procedure, top, intermediate and lower panels. When a top, intermediate or lower panel, Drawing 5979195, 5979196, or 5979197, is selected for testing, it shall be laid up and supported in the same manner as described for testing a side panel and all requirements for size, edge radii and length of supports and pressure head shall apply. The load of 12,000 ± 500 pounds shall be applied in eight, approximately equal, successive increments for the entire length of the panel as indicated on figure 4. The rate of load application shall be such that maximum load will occur in not less than three minutes nor more than five minutes. The maximum load shall be maintained for a period of not less than five minutes. The panel shall meet the requirements of 3.6.8 and shall be thoroughly suitable for the intended use.

4.5.4.4 Acceptance or rejection of inspection lot. The inspection lot shall be accepted if the panel selected and tested meets the requirements specified herein. Any panel failing to meet the requirements shall be cause for rejecting the inspection lot. The contracting activity reserves the right to retest any lot prior to acceptance.

4.5.5 Method B, destructive method. The lot shall be sampled, tested and accepted in accordance with 4.5.5.1 and 4.5.5.2. Method B shall be used for acceptance lot inspection only when there is a change in manufacturing processes during production of the lot.

4.5.5.1 Selection and preparation of sample. One panel shall be selected at random from the inspection lot, made with the new manufacturing process. The panel shall be subjected to the tests specified in 4.6.1 through 4.6.7.

4.5.5.2 Acceptance or rejection of inspection lot. The specimen shall meet the requirements for shear strength, flexural strength, heat resistance, density, and dimensional accuracy as specified in 3.6. Acceptance of inspection lot requires that all specimens pass the test specified herein. Failure of any specimen to meet the applicable requirements shall be cause for rejection of the inspection lot.

4.6 Test methods.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

4.6.1 Dimensional accuracy inspection. All specimens, prior to heating and undergoing shear and flexural testing, shall be measured to determine the conformance with requirements specified in 3.2.1. Specimens failing to meet the dimensions and tolerances specified on the drawings for the products covered by this specification shall be cause for rejection of the specimen for further inspection.

4.6.2 Heat resistance test. The test specimens to be used in the shear and flexural strength tests, prepared as specified in 4.6.3 and 4.6.4, shall be heated to a temperature of 168 + 0, -10°F. The specimens shall be examined for defects during the heating process and those which exhibit any of the defects specified in 3.6.1 shall be disqualified for further testing.

4.6.3 Shear strength test. Two pieces, 3.00 ± 0.12 inches wide, by 8.00 ± 0.12 inches long, shall be saw-cut from the sample for use as shear strength specimens. The ribbon direction shall run longitudinally. Support the test specimen in the manner shown on figure 1. Apply a force of 1680 + 40, - 0 pounds to pressure piece as shown on figure 1. The rate of load application shall be such that maximum loading will occur in not less than three minutes nor more than five minutes. The maximum load shall be maintained for a period of not less than five minutes. This test shall be performed immediately following the heat resistance test specified in 4.6.2, and while the specimens are still in the heated condition specified.

4.6.4 Flexural strength test. Two pieces, 3.00 ± 0.12 inches wide, by 48.00 ± 0.12 inches long, shall be saw-cut from the sample for use in the flexural strength test. The ribbon direction shall run either longitudinally or transversely. Support the test specimen in the manner shown on figure 2. Apply a force of 1320 + 30, -0 pounds (660 + 15 -0 pounds each support) to pressure piece P shown on figure 2. The rate of load application shall be such that maximum loading will occur in not less than three minutes nor more than five minutes. The maximum load shall be maintained for a period of not less than five minutes. This test shall be performed immediately following the heat resistance test specified in 4.6.2, and while the specimens are still in the heated condition specified.

4.6.5 Density. A number of the test specimens (as specified in the contract or purchase order (see 6.2)), cut from the sample and selected at the discretion of the Government inspector, shall be weighed and the density computed to verify conformance with the requirements specified in 3.6.4.

4.6.6 Lap shear test. Sections of the bonded edge extrusion of the sample shall be removed from randomly selected portions of the panel periphery and machined to form specimens in accordance with figure 3. Only two specimens shall be taken from any given four-foot section of the periphery. Six of the specimens shall contain the skin of one side of the panel and six of the specimens shall contain the skin of the opposite side of the panel. The twelve specimens shall be tested in the manner of the Normal Temperature Tensile Shear Test specified in MMM-A-132. Ten or more of the specimens shall test to a minimum bond shear strength of 2500 psi. Two specimens may test as low as 2000 psi, however, the average of all twelve specimens shall test to a minimum of 2500 psi.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

4.6.7 Short column compression test. Six specimens shall be cut from the sample at randomly selected locations such that at least two feet separate the locations. Each specimen shall be three inches wide by five inches long with the core ribbon direction parallel to the five inch dimension. Each specimen shall have the ends ground or machined flat and parallel to prevent premature failure. The six specimens shall be clamped in a suitable holding fixture and tested in the manner specified under the edgewise compression sandwich test of MIL-STD-401. The failing load average of the six specimens shall be 80,000 psi (30,500 pounds actual load) or greater with no two specimens less than 70,000 psi (28,500 pounds actual load).

4.7 Inspection of packaging. Packaging shall be inspected for conformance to the requirements of section 5.

5. PACKAGING

5.1 Packing. Packing shall be level A or C as specified in the contract or purchase order (see 6.2). Only like units shall be packed in the same packing crate. The total number of units in each crate shall be as specified in the contract or purchase order.

5.1.1 Level A. Panels shall be packed in wooden crates in accordance with MIL-C-104. Corrugated fiberboard separators and pads in accordance with PPP-B-636 Class I shall be provided as shown on figure 5. Case liners shall be provided in accordance with MIL-L-10547.

5.1.2 Level C. Panels shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any markings specified in contract or purchase order (see 6.2), shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers panels intended for use in the ASROC Launcher Mark 112.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. The types of panels required (see 1.2 and 3.2.1).

d. Whether a first article inspection is required (see 3.1).

e. Any materials not conforming to applicable drawings (see 3.3).

f. The agency that is to perform the inspection (see 4.4).

g. Identification of inspection lots and panels if required (see 4.5.1).

h. The number of test specimens to be cut for density test (see 4.6.5).

i. Whether level A or C packing is required and the total number of units to be packed in each crate (see 5.1).

j. Any marking other than as specified in 5.2.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definition, as referenced in 3.6.1, shall apply:

6.4.1 Permanent deformation. Permanent deformation shall be interpreted as deformations which arise from heating and testing, and which assume a permanent set in excess of the tolerances for flatness and parallelism of panel faces as specified on the applicable drawings.

6.5 Subject term (key word) listing.

Compression specimens
Lap shear specimens

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy - OS
(Project 5640-N554)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

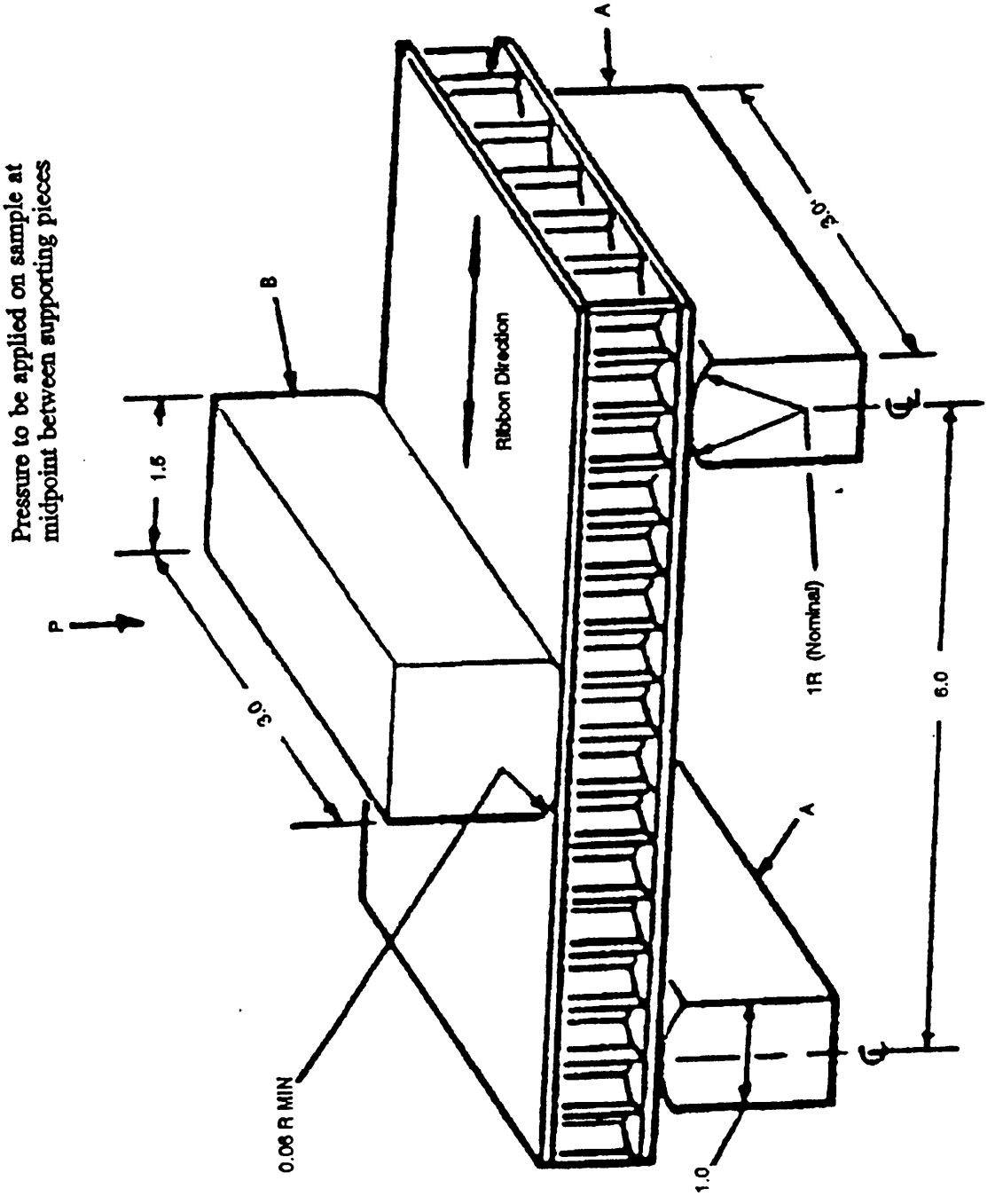


FIGURE 1. Shear strength test.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:34Z
Check the source to verify that this is the current version before use.

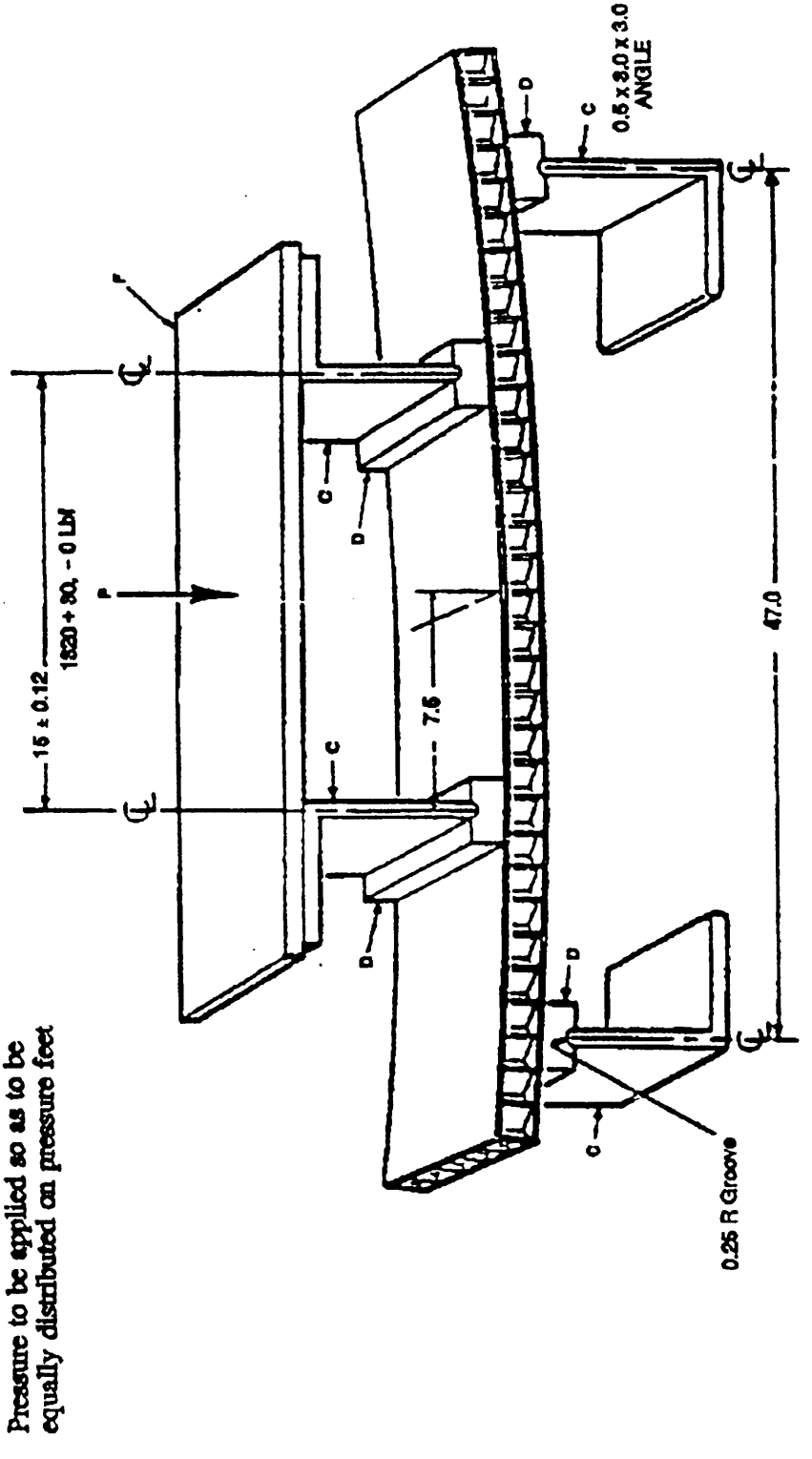


FIGURE 2. Flexural strength test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

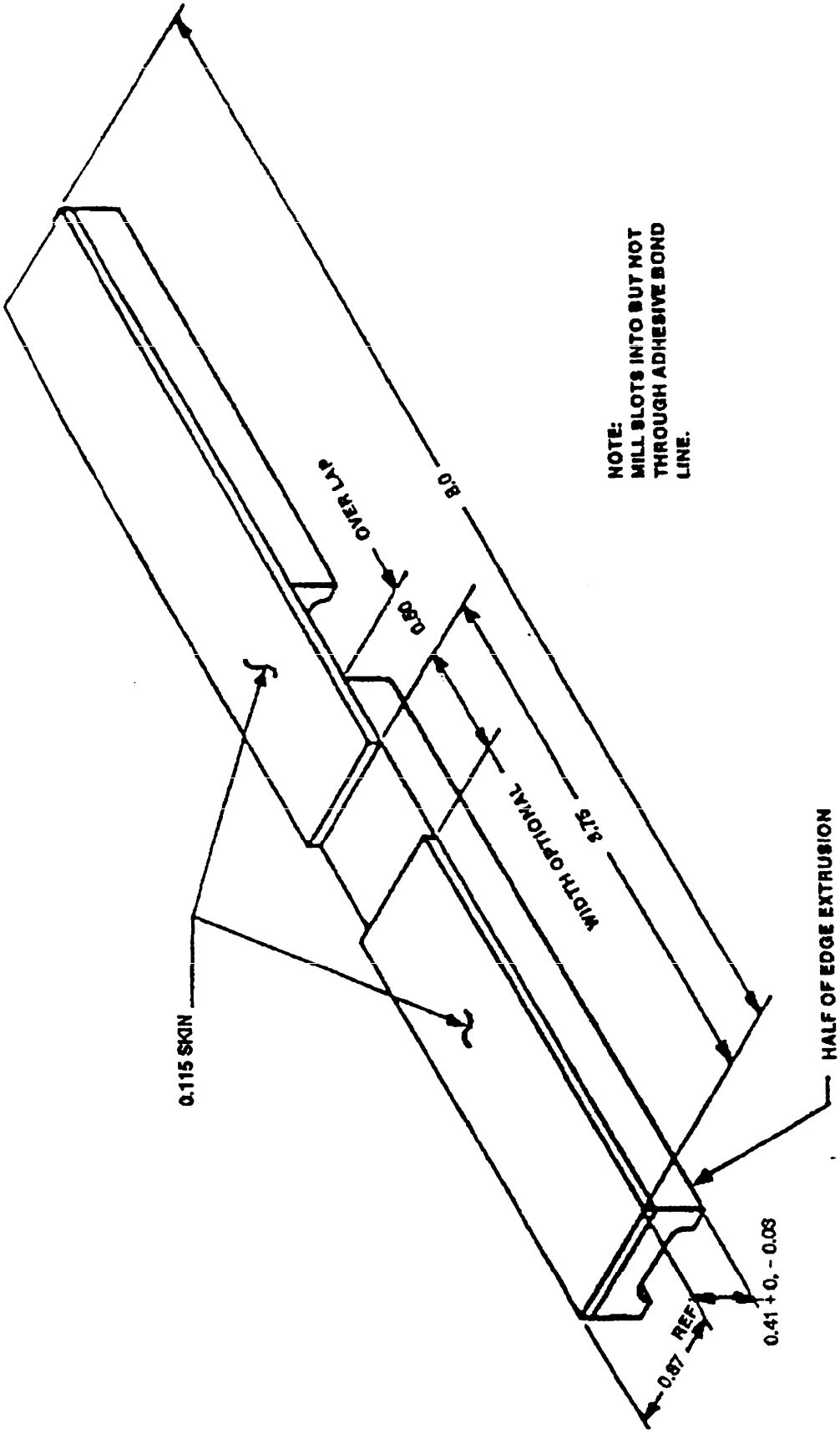


FIGURE 3. Lap shear test specimen.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

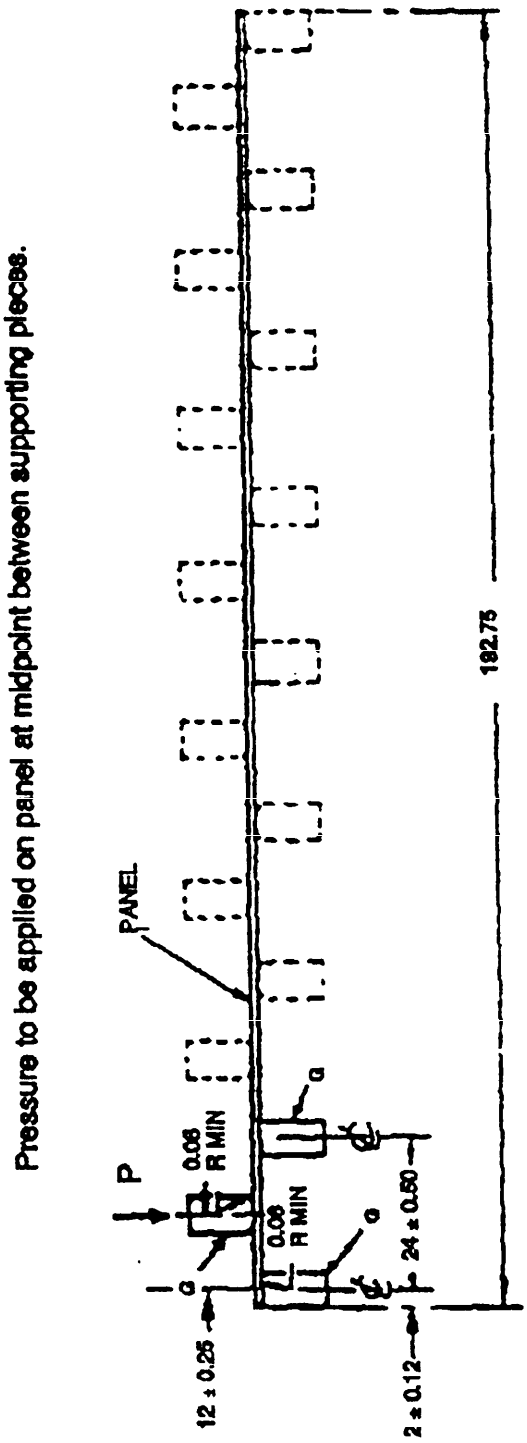


FIGURE 4. Method A nondestructive test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

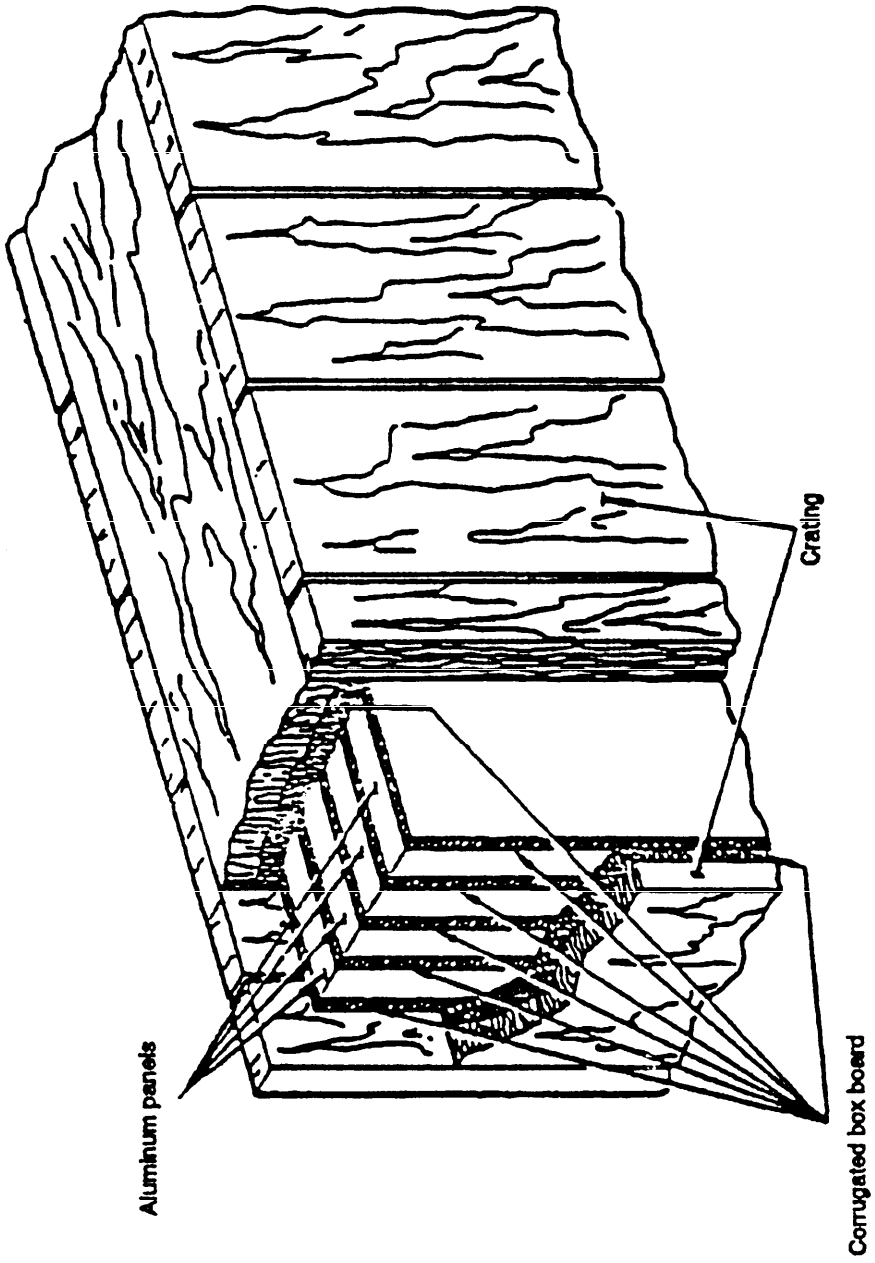


FIGURE 5. Packing, suggested method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-81293A(OS)	1992 JULY 22

3. DOCUMENT TITLE
LIGHTWEIGHT SANDWICH-CONSTRUCTED PANELS (for ASROC Launcher MK 112)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Commander, Indian Head Division Naval Surface Warfare Center (Code 0343)	(1) Commercial: 301-743-4510/4358 (2) AUTOVON: 364-4510/4358
c. ADDRESS (Include Zip Code) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.